

SIHYEON KIM

Master's Student in City and Regional Planning
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EDUCATION

Seoul National University, Graduate School of Environmental Studies	Seoul, Korea
M.S. in City and Regional Planning	2025 – Present
Soongsil University	Seoul, Korea
Bachelor of Architecture (B.Arch), 5-year program	2025

PUBLICATIONS

Peer-Reviewed Journal Articles

Quan, S. J.*, & Kim, S. (2025). Mixture of urban form archetypes in Seoul: A signal processing approach. *Urban Design International*.

Manuscripts in Preparation

Kim, S., Li, J., & Quan, S. J.* Comparing the cost-effectiveness of urban heat island mitigation strategies in Seoul's urban regeneration projects. Manuscript in preparation.

Kim, S., Li, J., & Quan, S. J.* Simulation-based optimization of heat-mitigation strategies under budget constraints for urban regeneration in Seoul. Manuscript in preparation.

* Corresponding author

CONFERENCE PRESENTATIONS

Kim, S. (2026). Evaluation of Cost-Effectiveness of Urban Heat Mitigation Strategies in Seoul's Regeneration Projects. 66th ACSF Annual Conference, United States, October 10–12.

Kim, S. (2026). Cost-Effective Heat Mitigation across Urban Regeneration Typologies: Implications for Urban Microclimate in Seoul. 20th IACP Conference, Xi'an, China, July 10–12.

Kim, S. (2026). Thermal Performance and Cost-Effectiveness of Urban Heat Mitigation Strategies in Seoul's Regeneration Projects. 2nd Workshop on AI for Urban Planning, AAAI-40, Singapore, January 26.

Kim, S. (2025). Assessing Urban Heat Mitigation Strategies in Seoul's Regeneration Projects. 19th IACP Conference, Xiamen, China, July 3–7.

Kim, S. (2025). Evaluating the Effectiveness of Heat Mitigation Strategies in Seoul's Urban Regeneration Projects. UDIK Spring Conference, Chuncheon, Korea, pp. 792–795.

Kim, S. (2024). Design Method of Incineration Plant Architecture as Healing and Recovery. Architectural Institute of Korea Spring Conference, Undergraduate Session, March 13.

RESEARCH EXPERIENCE

Seoul National University — Graduate School of Environmental Studies Seoul, Korea
Advisor: Professor Steven Jige Quan

Cost-Effective Urban Heat Mitigation across Seoul Urban Regeneration Typologies Jan 2025 – Present
First-Author Researcher, City Energy Lab

- Conducted the full research process, including statutory plan review, data collection, typology classification using principal component analysis and k means clustering, ENVI met model development and validation, and cost effectiveness analysis, to evaluate green roofs, bright roofs, and permeable pavement and develop planning recommendations for different urban regeneration typologies.

Simulation-Based Optimization of Heat-Mitigation Strategies under Budget Constraints for Urban Regeneration in Seoul Jan 2026 – Present

First-Author Researcher, City Energy Lab

- Designed and conducted the study by preparing spatial data, running SOLWEIG simulations, and applying a binary genetic algorithm to assess cooling performance, marginal returns, and spatial allocation outcomes under fixed budget.

Mixture of Urban Form Archetypes in Seoul: A Signal Processing Approach 2025 – In press

Second Author, City Energy Lab

- Led a qualitative historical Seoul's urban morphologies review of approximately 60 studies and connected SimplexNMF and clustering outputs to literature-based urban-form typologies through GIS overlays, a Sankey diagram, timelines, maps, and illustrations.

National Research Foundation of Korea (NRF) – AI and Data-Driven Design (AIDD): Sustainable Future City Design Integrating AI and Urban Big Data Sep 2025–Present

Research Assistant & Project Coordinator, NRF-Funded Research Project

- Coordinate project administration and budget management and prepared three grant applications, supporting research on sustainable future-city design through the integration of AI and urban big data.

UN-Habitat–Collaborative Smart Inclusive Transition Project for Growing and Shrinking Cities Jul 2025

Research Assistant, International Collaborative Research Project

- Collected, standardized, and analyzed multi-source data for six growing and shrinking cities in China to support the UN-Habitat–collaborative Smart Inclusive Transition Index and context-specific inclusive smart-city transition strategies.

TEACHING EXPERIENCE

Seoul National University — Graduate School of Environmental Studies Seoul, Korea

Advisor: Professor Steven Jige Quan

Teaching Assistant & Tutorials 2025, 2026

Courses: Zoning for Sustainable Development; Artificial Intelligence in Urban Planning and Design I & II; Urban Form and Energy

- Delivered two simulation tutorial sessions: building energy consumption analysis with DesignBuilder and urban microclimate simulation with ENVI-met.

International Education Experience

Chiles High School, Florida, USA | Grade 9 | 2015–2016

Deer Lake Middle School, Florida, USA | Grade 8 | 2014–2015

Meeker Elementary School, Iowa, USA | Kindergarten–Grade 1 | 2006–2007

HONORS & AWARDS

- Best Cross-disciplinary Integration Award, Participatory Urban Regeneration Empowered by Artificial Intelligence — 2nd AI4CITIES International Conference & Summer Workshop, Tongji University (2025).
- Departmental Honors Award, 3rd overall, Soongsil University (2025).
- Graduation Design Award, 4th place, Soongsil University (2024).
- Encouragement Award, Modular Mobility Urban Regeneration Project, Korea Space Design Association National Competition (2023).
- Award Winner, Eco-friendly Office Building Design, Korean Society of Basic Design & Art; Sejong Center Exhibition (2023).
- Three-Time Award Winner, JoongAng Painting Competition; selected works exhibited at the Korea Art Museum (2022–2024)
- First Prize, AP Art Awards; City Hall Winter Youth Art Exhibition, Tallahassee, Florida (2014,2015)
- Special Prize in Fine Art, 5th Art Korea World Art Competition (2026)
- Award Winner, Korean Women's Painting Competition (2026)
- Award Winner, Oil Painting Division, Korean Society of Basic Design & Art; exhibited at the Sejong Center (2023)

PROFESSIONAL EXPERIENCE

Design Somnium Seoul, Korea

Architecture Intern — GCC Multi-purpose Cultural Center Remodeling Project Jan 2024 – Jul 2024

- Prepared daily construction reports and documented demolition, structural reinforcement, façade, and interior work.
- Supported design–construction coordination through CAD and SketchUp modeling and on-site communication.

SKILLS

Programming & Data Analysis: Python, statistical analysis, machine learning, genetic algorithms, optimization

GIS & Urban Analytics: ArcGIS Pro, QGIS, raster/vector processing, spatial databases

Environmental Simulation & Optimization: ENVI-met, SOLWEIG, DesignBuilder, IES VE

Design & Visualization: AutoCAD, SketchUp, Rhino 3D, Grasshopper, Ladybug Tools, V-Ray, Photoshop, Illustrator, InDesign, Figma

CERTIFICATIONS & LANGUAGES

Rhino Certification (2025.08) | ESG SNU Youth Forum Certification (2026)

Korean: Native | English: Professional working proficiency

ADDITIONAL EXPERIENCE

- Sign Language Interpreter Volunteer — Real-time interpretation for church services (2019–Present).

ARCHITECTURAL DESIGN PROJECTS

Soongsil University — School of Architecture, College of Engineering

Seoul, Korea

Individual academic projects; independently completed concept development, site analysis, architectural design, drawings, diagrams, and final visualization.

Healing Fragmentation: Mokdong Incinerator Regeneration

Graduation Project, 2024

Graduation Design Award (4th Place), Soongsil University

- Reframed an essential but socially isolated waste facility as a mediating public landscape connecting people, nature, and industrial systems; developed a phenomenological healing sequence through changing viewpoints, movement, light, and adaptive reuse.

Urban Play Hub: Yongsan Electronics Market Regeneration

Urban Design Studio, 2023

- Used AI-assisted massing exploration to regenerate a declining electronics district around productive play, healing, logistics, and community programs; reorganized vehicle, pedestrian, and freight circulation at the master-plan scale.

Circulation of Communication: Mobility-Based Yongsan Renewal

Group Competition Project, 2023

Encouragement Award, Korea Space Design Association National Competition

- Proposed PBV mobility units, modular mixed-use additions, a pedestrian linear plaza, and separated underground logistics to demonstrate future-oriented urban design responsive to changing retail, work, and housing patterns.

Smart Farm + Skin: Eco-Friendly Office

Environmental Design Studio, 2022

Award Winner, Korean Society of Basic Design & Art

- Integrated smart farms, terraces, and a double-skin facade; used IES VE to compare perforation, louver, and window-size alternatives for daylight, glare control, and reduced lighting-energy demand.

Connecting Multi Paths into Residents' Heart

Housing Studio, 2020

Outstanding Architectural Design Award (Top Rank in Class)

- Designed a stepped artist-housing community in Huam-dong that preserved local topography and urban scale while reconnecting fragmented green space through shared courtyards, private gardens, ateliers, and extended alleys.

Soyo-yu (逸遙遊): Museum Remodeling and Structural Reuse

Adaptive Remodeling Studio, 2021

- Preserved the primary core and columns while creating a vertical void, ramps, sunken spaces, and a sequence of metal, wood, and light galleries to reinterpret the work and cultural identity of architect Itami Jun.